

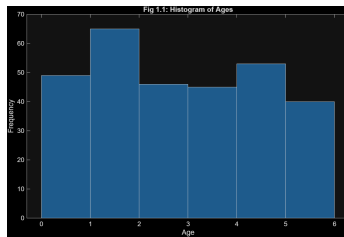
Homework 1 ECO351

Palak Vadher (230728)

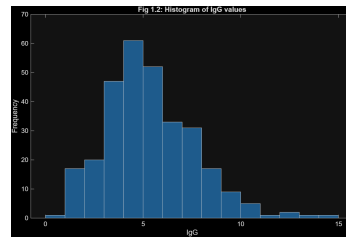
		mean	median	std	max	min
1. (a)	Age	2.774	2.583	1.665	6.000	0.500
	IgG	5.292	5.000	2.277	14.400	0.900

Table 1: Summary of Age and IgG

The following are the histograms for Age and IgG



(a) Histogram for Age



(b) Histogram for IgG

$$Skewness = \frac{\sum_{i=1}^N (X_i - \bar{X})^3}{(N-1)\sigma^3} \quad (1)$$

By calculating skewness, we see that skewness(age)=0.2662 and skewness(igg)=0.7685. Hence, age is right-skewed but very slightly. However, IgG is moderately right-skewed as its skewness is much closer to +1.

(b) *Model* : $y_i = \beta_1 + \beta_2 x + u_i$

	Estimate	SE	t-stats	p-values
Intercept	3.364	0.2214	15.194	6.097e-39
Age	0.6951	0.0684	10.152	5.659e-21

Table 2: Estimations for Model 1

R-squared= 0.258 or 25.8%.

Here, $\beta_{age} = 0.6951 > 0$ which means that IgG and Age are positively correlated. Moreover, the p-value for age is also less than 0.001 which leads us to infer statistical significance in our results. Hence, the null hypothesis $H_0 : \beta_{age} = 0$ is rejected.

(c) *Model* : $y_i = \beta_1 + \beta_2 x + \beta_3 x_i^2 + \varepsilon_i$

	Estimate	SE	t-stats	p-values
Intercept	3.0839	0.3833	8.0458	2.108e-14
Age	0.9674	0.3118	3.1029	2.102e-3
Agesqr	-0.0454	0.0507	-0.8954	0.3713

Table 3: Estimations for Model 2

R-squared= 0.26 or 26%.

For 'age', we take $H_0 : \beta_{age} = 0$ and $H_a : \beta_{age} \neq 0$. We see that the estimate is positive. This implies a positive correlation. Also, t-stat for age shows statistical significance as t-stat= 3.1029 > 2. The p-value is also very small, indicating that the data provides enough evidence to reject the null hypothesis.

For 'agesqr', we take $H_0 : \beta_{agesqr} = 0$ and $H_a : \beta_{agesqr} \neq 0$. We see that the estimate $\hat{\beta}_{agesqr}$ is negative, implying a negative correlation between agesqr and IgG. However, the value of the t-statistic lies between the critical values of approximately -2 and +2 which means we cannot reject the null hypothesis. The p-value is also greater than 0.05, meaning 'agesqr' is not statistically significant.

(d) *Model2* : $y_i = \beta_1 + \beta_2 x + \beta_3 x_i^2 + \varepsilon_i$

The Likelihood function is as follows:

$$L(y_i, x_i, \theta) = \prod_{i=1}^n f(y_i, x_i, \theta)$$

We have been given that error terms follow a standard normal distribution. Hence, we substitute the probability distribution function into the likelihood function.

$$L = \prod_{i=1}^n \frac{e^{-\frac{(y_i - \beta_1 - \beta_2 x - \beta_3 x_i^2)^2}{2}}}{\sqrt{2\pi}} \quad (2)$$

Taking log, we get the log-likelihood function

$$LL(\beta_1, \beta_2, \beta_3) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x - \beta_3 x_i^2)^2 \quad (3)$$

	Estimate	SE
Intercept	3.0839	0.3833
Age	0.9674	0.3118
Agesqr	-0.0454	0.0507

Table 4: Estimations for Model 2 using Newton-Raphson method

We see that the estimates calculated using the Newton-Raphson algorithm are equal to the ones calculated using the in-built estimation functions in MATLAB.

		Count	Percent
2. (a)	No, illegal (0)	581	46.85%
	Yes, legal (1)	659	53.15%

Table 5: Data Summary for y

	mean	std dev
(b) age	50.501	17.758
hh1	2.719	1.440

Table 6: Data Summary for age and hh1

	Count	Percent
(c) No(0)	653	52.66%
Yes(1)	587	47.34%

Table 7: Data Summary for past use

(d) For 'sex',		Count	Percent
	Male	603	48.63%
	Female	637	51.37%

Table 8: Data Summary for 'sex'

For 'parent',		Count	Percent
	Yes	358	28.87%
	No	882	71.13%

Table 9: Data Summary for 'parent'

		Count	Percent
(e)	post-married	285	22.98%
	couple	720	58.06%
	single	235	18.95%

Table 10: Data Summary for 'marital status'

		Count	Percent
(f)	poor	642	51.77%
	rich	232	18.71%
	middle	366	29.52%

Table 11: Data Summary for 'income'

		Count	Percent
(g)	lessThanBach	381	30.73%
	BachelorsandAbove	445	35.89%
	HSandBelow	414	33.39%

Table 12: Data Summary for 'educ'

		Count	Percent
(h)	white	954	76.94%
	black	146	11.77%
	allOther	140	11.29%

Table 13: Data Summary for 'race'

		Count	Percent
(i)	democrat	422	34.03%
	republican	357	28.79%
	independentOthers	461	37.18%

Table 14: Data Summary for 'party'

3. (a) The model is given as

$$z_i = x_i' \beta + \epsilon_i, \epsilon_i \sim N(0, 1)$$

$$y_i = \begin{cases} 1 & , z_i > 0 \\ 0 & , z_i \leq 0 \end{cases}$$

Hence,

$$Pr(y_i = 0) = Pr(\epsilon_i \leq -x_i' \beta) = \Phi(-x_i' \beta) \quad (4)$$

Moreover, $Pr(y_i = 1) = 1 - Pr(y_i = 0)$, which in this case becomes easy as we have a standard normal distribution.

$$Pr(y_i = 1) = 1 - \Phi(-x_i' \beta) = \Phi(x_i' \beta) \quad (5)$$

The likelihood function is given as $\prod_{i=1}^n f(y_i|\beta)$

$$\mathcal{L}(\beta) = \prod_{i=1}^n \Phi(x'_i\beta)^{(y_i)} [1 - \Phi(x'_i\beta)]^{(1-y_i)} \quad (6)$$

		Estimate	SE	t-stats
	Intercept	0.676	0.093	7.279
	age	-0.003	0.001	-3.106
	hh1	-0.021	0.011	-1.870
	past use	0.292	0.027	10.743
	sex	0.056	0.027	2.067
	parent	0.018	0.0362	0.499
	single	0.040	0.042	0.968
(b)	post-married	0.018	0.037	0.497
	poor	-0.062	0.041	-1.522
	middle	-0.059	0.039	-1.521
	HSandBelow	-0.062	0.034	-1.818
	lessThanBachelors	0.024	0.034	0.717
	white	-0.016	0.043	-0.391
	black	-0.110	0.055	-2.002
	democrat	0.077	0.032	2.396
	republican	-0.166	0.033	-5.041

Table 15: Estimation using probit model

- (c) The change in probability $Pr(y_i = 1|\beta)$ observed when Age is changed to (Age+5) is -0.0054. This means that when age increases by 5 years, the probability of saying 'Yes, legal' decreases by 0.0054.
- (d) The change in probability $Pr(y_i = 1|\beta)$ observed is 0.0062. In other words, the probability of supporting legalisation increases by 0.0062 when the individual switches from non-parent to parent.

4. (a) The model is given as $z_i = x'_i\beta + \epsilon_i, \epsilon \sim \mathcal{L}(0, 1)$

$$y_i = \begin{cases} 0 & , z_i \leq 0 \\ 1 & , otherwise \end{cases}$$

Hence,

$$Pr(y_i = 1) = 1 - Pr(\epsilon_i \leq -x'_i\beta) = 1 - \Lambda(-x'_i\beta) = \Lambda(x'_i\beta) \quad (7)$$

Following from the rules of probability,

$$Pr(y_i = 0) = 1 - \Lambda(x'_i\beta) \quad (8)$$

The likelihood function is given as:

$$\mathcal{L}(\beta) = \prod_{i=1}^n \Lambda(x'_i\beta)^{(y_i)} [1 - \Lambda(x'_i\beta)]^{(1-y_i)} \quad (9)$$

		Estimate	SE	t-stats
	Intercept	0.857	0.442	1.934
	age	-0.015	0.004	-3.047
	hh1	-0.100	0.055	-1.814
	past use	1.304	0.130	10.032
	sex	0.266	0.130	2.051
	parent	0.078	0.173	0.451
	single	0.195	0.201	0.974
(b)	post-married	0.085	0.176	0.486
	poor	-0.298	0.199	-1.498
	middle	-0.281	0.189	-1.484
	HSandBelow	-0.296	0.166	-1.783
	lessThanBachelors	0.122	0.164	0.748
	white	-0.069	0.207	-0.336
	black	-0.518	0.264	-1.963
	democrat	0.366	0.155	2.360
	republican	-0.783	0.158	-4.938

Table 16: Estimation using logit model

- (c) The covariate effect of age when age is increased by 5 years, calculated by the logit model, comes out to be -0.0153. We can say that an individual aging 5 more years decreases the probability of them supporting legalisation of marijuana by 0.0153.
- (d) The covariate effect of being a parent is observed to be 0.0162. A unit change in the variable 'parent' from 0 to 1 increases the probability of supporting marijuana legalisation by 0.0162.
- (e) Yes, there is a difference in the covariate effects calculated using the probit and logit models. The covariate effects calculated by

the logit models are greater in magnitude than those by the probit model. This is because the Logit model has fatter tails which means that it is quite sensitive to outliers. Hence, even a slight change in their attributes can cause a more significant change or the logit model than the probit model.

5. (a) For an ordinal probit model, the $Pr(y_i = j)$ is given by

$$Pr(y_i = j) = \Phi(\gamma_j - x'_i\beta) - \Phi(\gamma_{j-1} - x'_i\beta) \quad (10)$$

The likelihood function is calculated as:

$$\mathcal{L}(\beta, \gamma; y) = \prod_{i=1}^n \prod_{j=1}^J Pr(y_i = j | \beta, \gamma)^{I(y_i=j)} = \prod_{i=1}^n \prod_{j=1}^J [\Phi(\gamma_j - x'_i\beta) - \Phi(\gamma_{j-1} - x'_i\beta)]^{I(y_i=j)} \quad (11)$$

- (b) Table 17: Summary of continuous and categorical variables

	Count/Mean	Percent/Std
educLevel 1	897	22.9%
educLevel 2	1392	35.5%
educLevel 3	876	22.3%
educLevel 4	758	19.3%
famIncome	28.975	22.362
educMother	10.877	3.093
educFather	10.859	3.819
motherWorked=0	1759	44.84%
motherWorked=1	2164	55.16%
female=0	1998	50.93%
female=1	1925	49.07%
black=0	3023	77.06%
black=1	900	22.94%
urban=0	843	21.49%
urban=1	3080	78.51%
south=0	2528	64.44%
south=1	1395	35.56%
age15=0	2813	71.71%
age15=1	1110	28.29%
age16=0	2762	70.41%
age16=1	1161	29.59%
age17=0	2908	74.13%
age17=1	1015	25.87%

	Estimate	SE	t-stats
Intercept 1	1.350	0.093	14.375
Intercept 2	2.438	0.097	25.019
Intercept 3	3.194	0.101	31.729
sqrt(famIncome)	-0.140	0.010	-13.625
educMother	-0.050	0.007	-6.519
educFather	-0.072	0.006	-11.516
(c) motherWorked	-0.032	0.036	-0.887
female	-0.164	0.035	-4.691
black	-0.152	0.044	-3.441
urban	0.043	0.043	1.006
south	-0.055	0.038	-1.442
age15	0.027	0.054	0.498
age16	-0.005	0.054	-0.094
age17	-0.230	0.056	-4.101

Table 18: Estimation using ordinal probit model

The Hit-Rate of the model comes out to be 0.435 or 43.5%. This means that the model predicts success when there actually is success, 43.5% of the time.

The McFadden's R-squared is 0.0885.

- (d) Average effect of a \$10,000 increase in family income:
Change in probability of "1": -0.2285 or -22.85%
Change in probability of "2": -0.3512 or -35.12%
Change in probability of "3": -0.2261 or -22.61%
Change in probability of "4": +0.8058 or +80.58%
Hence, we observe that when given \$10,000, the probability of the individual attaining a college or graduate degree increases by 80.58%.